
Reference Free Grading of Reasoning Degrades on Problems Inside the Instella Pretraining Corpus

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Abstract

1 Automated evaluators are trained artifacts, so a verdict on a benchmark item may
2 depend on whether the evaluator encountered that item during training, yet the cor-
3 pora of models deployed as judges are closed and the question is normally left open.
4 Instella publishes its complete pretraining mixture and its stage two component
5 derives from the GSM8K training split rather than the test split, so membership
6 becomes a measured property fixed by exact thirteen gram containment instead of
7 an estimated one. Grading 2017 candidate solutions produced by Instella check-
8 points, five judges spanning Instella 3B, Llama 3.1 8B and Qwen2.5 at 7, 14 and
9 32 billion parameters show a membership gap in balanced accuracy that depends
10 almost entirely on whether the gold answer accompanies the prompt. Supplied with
11 the reference the gap runs from -0.023 to -0.074 and shrinks as judges improve,
12 reaching -0.023 $[-0.057, +0.006]$ at 14 billion parameters. Withheld, it runs
13 to -0.243 $[-0.282, -0.202]$ and does not close with scale, standing at -0.191
14 $[-0.238, -0.140]$ for the largest judge tested. Decomposition locates the effect
15 entirely in specificity: sensitivity gaps span zero at every judge, at most $+0.011$
16 $[-0.005, +0.035]$, while specificity falls by -0.496 $[-0.572, -0.416]$, meaning
17 Llama 3.1 8B rejects 52.2 percent of wrong solutions to unseen problems and 2.7
18 percent of wrong solutions to seen ones. Familiarity produces credulity rather than
19 confusion, which falsifies an account based on comparison against a remembered
20 answer since that predicts a sensitivity loss. The effect survives a paraphrased
21 judging prompt, -0.254 $[-0.293, -0.215]$, and weakens only where candidate
22 errors are blunt enough to catch without a reference, -0.035 $[-0.077, +0.007]$
23 for the strongest judge on the weakest generator. Cohen’s kappa computed on
24 identical verdicts reports -0.485 where balanced accuracy reports -0.243 , an
25 inflation traceable to a base rate difference of 0.772 against 0.526 rather than to any
26 change in agreement, so chance corrected statistics should not size membership
27 effects when arms differ in base rate. The configuration in which the gap is largest,
28 grading without a gold answer, is exactly the configuration deployed whenever no
29 gold answer exists.

30 1 Introduction

31 An LLM judge is asked to certify reasoning it may have been trained on. Whether that matters is an
32 empirical question that is normally unanswerable, because the pretraining corpora of models used
33 as judges are not published, so membership has to be estimated from proxies such as perplexity or
34 benchmark contamination heuristics. Instella [Liu et al., 2025] removes that obstacle. Its complete
35 pretraining mixture is released, and the stage two synthetic component is generated from the GSM8K
36 training split [Cobbe et al., 2021] and not from the test split. Membership therefore becomes a
37 property that can be measured on each item by exact thirteen gram containment rather than inferred,
38 which turns a question about contamination into a controlled contrast.

39 The design holds everything fixed except one binary variable. Identical benchmark items, identical
40 candidate solutions and identical judges are graded twice, once with the gold answer supplied and
41 once without. Reliability is scored as balanced accuracy against exact match ground truth, and the
42 quantity of interest is the seen minus unseen difference within each condition. Any property of the
43 items themselves, their difficulty, their phrasing or their length, enters both conditions equally and
44 cancels.

45 Three findings follow. Reliability loss on training corpus problems is small and closes with capability
46 when the reference is supplied, and large and persistent when it is withheld. The loss is a specificity
47 failure and not a sensitivity failure, so judges become credulous about familiar problems rather than
48 confused by them, which rules out the intuitive account in which a judge compares a candidate
49 against a remembered answer. Chance corrected agreement inflates the effect substantially on
50 identical verdicts, for reasons that follow from the marginals and not from the judging.

51 **Related work.** Judge reliability is normally studied through agreement with human raters [Zheng
52 et al., 2023]. Contamination work runs in parallel and asks whether a benchmark score is inflated
53 by training exposure, either by detecting membership from model behaviour [Shi et al., 2024], by
54 rephrasing items so that surface overlap fails [Yang et al., 2023], or by rebuilding a benchmark from
55 scratch to see how much of a reported score survives [Zhang et al., 2024]. That literature almost
56 always concerns the model under test rather than the model doing the grading, and memorisation is
57 quantified for generation rather than for judgement [Carlini et al., 2023]. The contribution here sits at
58 the join of the two. Membership is fixed by a published corpus rather than estimated from behaviour,
59 and the quantity measured is the evaluator’s reliability rather than the generator’s score.

60 2 Design

61 **Membership.** Containment is computed as the fraction of an item’s thirteen grams appearing in
62 a single corpus document, taking the maximum over documents. The stage two subset the model
63 actually consumed, `train_119K`, holds 119,014 rows. Calibration fixes the threshold: against the full
64 released pool of 1,367,882 rows, no GSM8K test item reaches containment 0.50 and the maximum
65 observed is 0.487, confirming the corpus is train derived and that the train against test partition is a
66 genuine exposure contrast rather than a difficulty contrast.

67 **Candidate solutions.** Instella checkpoints generate solutions to both arms under identical decoding.
68 Ground truth is exact match against the reference answer. Base rates differ between arms by
69 construction, 0.772 correct on the seen arm against 0.526 on the unseen, which is why balanced
70 accuracy rather than raw agreement is the primary statistic.

71 **Judging.** Each judge grades every solution twice. The reference given condition places the gold
72 answer in the prompt; the reference withheld condition is identical in every other respect. Verdicts
73 that fail to parse are excluded throughout, 3 of 2017 and 17 of 2017 for the largest judge.

74 **Inference.** Intervals are cluster bootstraps over parent problems with 4000 draws at seed 6198,
75 because several graded variants can descend from one problem and resampling rows rather than
76 problems would understate the interval.

77 3 Results

78 3.1 The gap depends on whether a reference is supplied

79 Table 1 gives the membership gap for every judge under both conditions. Supplied with the gold
80 answer, the gap is small and shrinks as judges improve, reaching -0.023 $[-0.057, +0.006]$ at 14
81 billion parameters, an interval containing zero. Withheld, the gap is three to eight times larger at
82 every judge and shows no tendency to close: the 32 billion parameter judge, whose reference free
83 balanced accuracy on the seen arm is the highest measured at 0.648, still carries a gap of -0.191
84 $[-0.238, -0.140]$.

85 A null or an effect from an incompetent judge would say nothing, so absolute performance is reported
86 alongside the gaps. Figure 2 shows both. Given the reference, the 7, 14 and 32 billion parameter

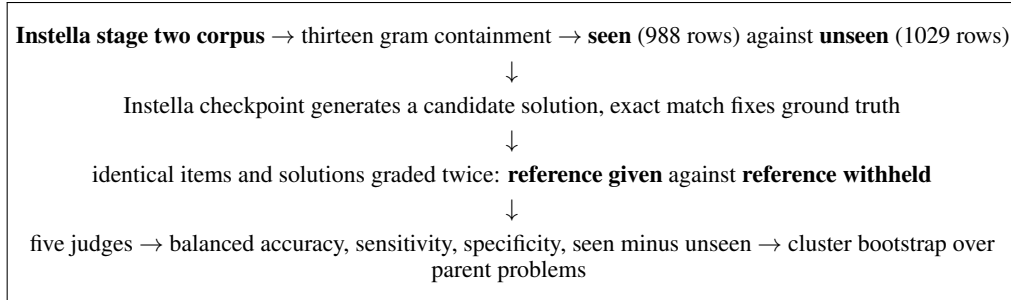


Figure 1: Membership is fixed by containment against a published corpus, candidate solutions are held identical across conditions, and the single manipulated variable is whether the gold answer accompanies the judging prompt.

Table 1: Membership gap in balanced accuracy, seen minus unseen, grading instruct outputs. Intervals are cluster bootstraps over parent problems. An asterisk marks an interval excluding zero. The ratio column is the reference free gap divided by the control gap.

Judge	Reference given (control)	Reference withheld	Ratio
Instella 3B Instruct	−0.054 [−0.083, −0.026]*	−0.081 [−0.109, −0.054]*	1.5
Llama 3.1 8B Instruct	−0.074 [−0.116, −0.032]*	− 0.243 [−0.282, −0.202]*	3.3
Qwen2.5 7B Instruct	−0.036 [−0.068, −0.008]*	− 0.200 [−0.248, −0.149]*	5.6
Qwen2.5 14B Instruct	−0.023 [−0.057, +0.006]	− 0.184 [−0.231, −0.133]*	8.0
Qwen2.5 32B Instruct	−0.062 [−0.103, −0.022]*	− 0.191 [−0.238, −0.140]*	3.1

judges reach 0.888 to 0.973 balanced accuracy, which is the regime in which their reference free numbers become interpretable. The 3 billion parameter judge sits within 0.04 of chance on the seen arm in every configuration, including when handed the answer, so its rows describe a noise floor and the substantive claims rest on the larger three. The two control series converge as capability rises while the two reference free series stay apart, which is the dissociation stated as a picture.

3.2 The failure is specificity, not sensitivity

Balanced accuracy averages two rates that can move independently, and separating them identifies the mechanism. Figure 3 decomposes both conditions. Sensitivity, the rate of accepting correct solutions, carries no membership gap anywhere: +0.011 [−0.005, +0.035] for Llama 3.1 8B and −0.004 [−0.024, +0.016] for Qwen2.5 32B, both spanning zero. Specificity, the rate of rejecting wrong solutions, collapses: −0.496 [−0.572, −0.416] and −0.378 [−0.471, −0.281] respectively. In absolute terms Llama 3.1 8B rejects 52.2 percent of wrong solutions to unseen problems and 2.7 percent of wrong solutions to seen ones.

That asymmetry falsifies the account the design was originally built to test. A judge comparing a candidate against a remembered answer would mark correct solutions wrong whenever its memory were imperfect, producing a sensitivity loss. No such loss appears at any judge in either condition. What appears instead is credulity: familiarity with a problem raises the probability that a wrong solution to it is waved through.

3.3 Chance corrected agreement inflates the effect

Cohen’s kappa is the reflex choice for judge reliability and it misleads here. Figure 4 runs both statistics over one set of verdicts. With the reference supplied the two agree to three decimals, both reporting −0.074. With it withheld, kappa reports −0.485 where balanced accuracy reports −0.243, roughly double.

Nothing about agreement changed to produce that. Ground truth base rates are identical across the two conditions, 0.772 on the seen arm against 0.526 on the unseen, because the graded solutions do not change. What moves is the judge’s own verdict marginal, and kappa depends on both marginals as well as on agreement. An arm near one half carries more chance corrected headroom than a skewed arm, so skew alone shifts the statistic. Feinstein and Cicchetti [Feinstein and Cicchetti, 1990]

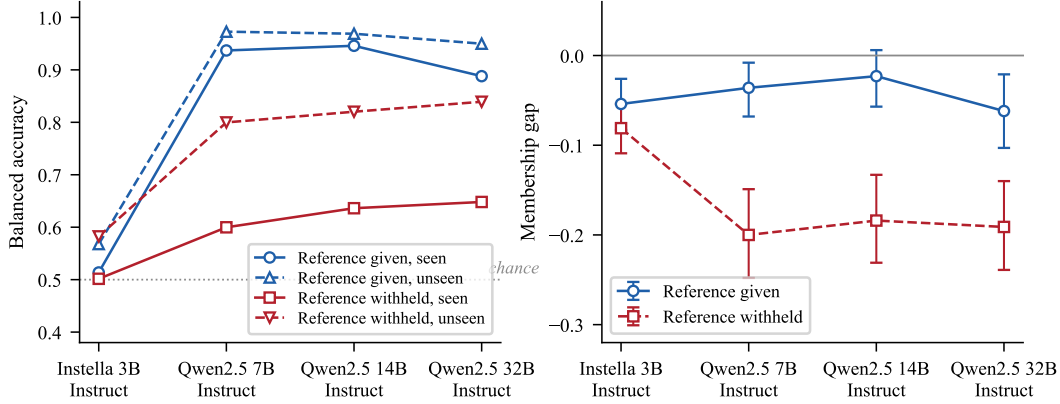


Figure 2: Left: absolute balanced accuracy for all four cells across the capability ladder. Right: the membership gap with cluster bootstrap intervals. Capability lifts every series and closes the control gap, leaving the reference free gap intact.

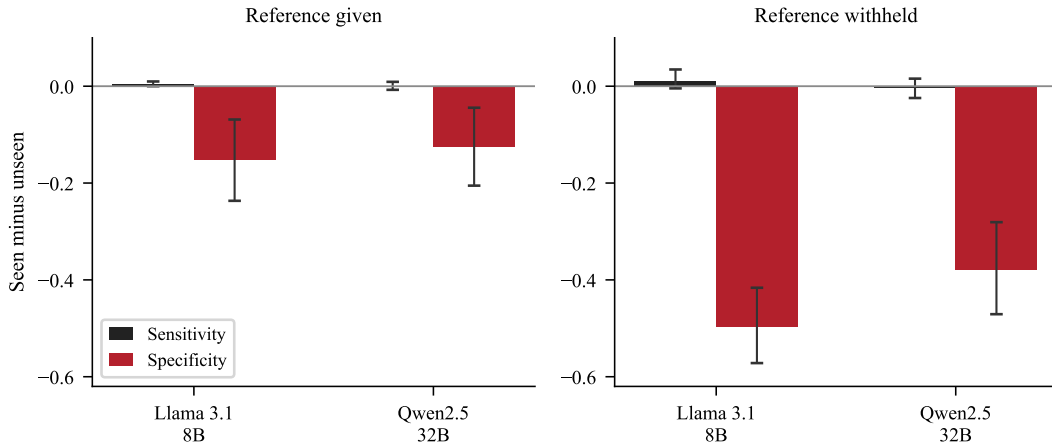


Figure 3: Sensitivity and specificity gaps, seen minus unseen, under both conditions. Sensitivity intervals contain zero everywhere. Specificity falls sharply once the reference is withheld, which is the entire effect.

115 describe this as the first kappa paradox. The practical consequence for evaluator reporting is narrow
 116 and concrete: where arms differ in base rate, a chance corrected statistic will size a membership
 117 effect wrongly even when it points the right way.

118 3.4 Robustness

119 Two explanations compete with the account above and both are testable. Figure 4 addresses them.
 120 A paraphrased judging prompt, rewritten to preserve the task and change the wording, leaves the
 121 effect intact and slightly larger: -0.254 $[-0.293, -0.215]$ for Llama 3.1 8B against -0.243 with the
 122 primary prompt, and -0.219 $[-0.261, -0.172]$ for Qwen2.5 32B against -0.191 . Prompt wording
 123 does not drive it.

124 Varying the generator does matter, and in the direction the account predicts. Grading the weaker stage
 125 two checkpoint, whose errors are blunter, the strongest judge shows -0.035 $[-0.077, +0.007]$, an
 126 interval containing zero, while Llama 3.1 8B on the same target still shows -0.192 $[-0.232, -0.151]$.
 127 Errors coarse enough to catch without a reference leave nothing for familiarity to erode, so the effect
 128 has a ceiling in how gradeable the candidate solutions are.

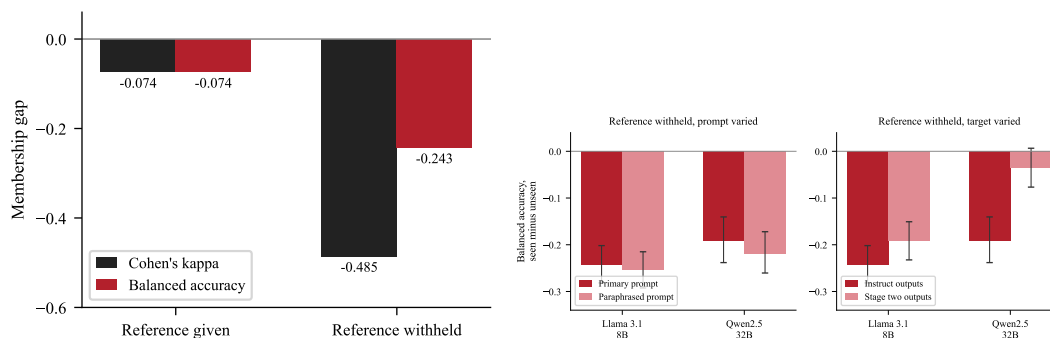


Figure 4: Left: two statistics on identical verdicts for Llama 3.1 8B. They agree under the control condition and diverge by a factor of two once the reference is withheld. Right: the reference free gap under a paraphrased judging prompt and against a weaker generator.

4 Implications for evaluator construct validity

A judge is deployed as an instrument measuring solution correctness. The results above say that instrument's specificity carries a term that depends on whether the graded item was in its training corpus, and that the term is nearly invisible whenever a gold answer is available. Benchmark practice makes this awkward, because the two conditions are not equally common. Reference free grading is used precisely where no gold answer exists, on open ended generation, on tasks scored by preference, and on new items for which references have not been written. Those are the settings in which the measured degradation is largest.

The size is operationally relevant rather than marginal. A judge rejecting 52.2 percent of wrong answers on unfamiliar problems and 2.7 percent on familiar ones is not applying one standard with noise. It is applying two standards, and which one applies is determined by a property of the judge's training data that the evaluation does not record and usually cannot observe. Reporting balanced accuracy split by sensitivity and specificity, rather than a single agreement number, makes the failure visible at low cost.

5 Limitations

Membership is verified for one judge, and it is not the judge carrying the result. Containment is measured against the Instella corpus, so the Instella judge has verified membership but sits at chance. The Qwen and Llama judges show the effect and their corpora are unknown, so for them the partition is GSM8K train against test rather than verified seen against unseen. That asymmetry is why the primary claim is stated as a within judge contrast between two conditions over identical items and identical candidate solutions, which holds whatever the arm label means for a given judge. Closing the gap to a membership claim requires a second open corpus model of comparable capability.

One family supplies the candidate solutions. Every graded output comes from an Instella checkpoint, and whether the dissociation reproduces on other generators is untested. The judges span three families.

The effect has a ceiling in solution quality. The stage two result above is measured on one generator pair and should be treated as an observation rather than a law.

One checkpoint is unexplained. Stage one outputs are wrong nearly everywhere and trivially gradeable in either condition, which accounts for that null. The supervised fine tuned checkpoint shows no gap in either condition despite output quality comparable to instruct, and the present data do not explain why.

Attribution is not established. No evidence links a verdict to a specific training document. Section 3.2 narrows the effect to a specificity loss, which is not attribution.

6 Conclusion

Measured against membership fixed by containment rather than estimated, an LLM judge shows a small reliability loss on problems from its training corpus when the reference answer is supplied, and a large one when it is withheld. The gap under reference free grading runs to -0.243 in balanced accuracy and does not close with scale, while the same judges in the control condition converge toward zero as they improve. The loss is entirely a specificity failure, so familiarity buys credulity rather than confusion. Chance corrected agreement roughly doubles the apparent effect on identical verdicts and should not be used to size it where arms differ in base rate. The operational reading is specific: reference free grading is the configuration deployed wherever no gold answer exists, and it is the configuration in which the items most likely to sit in a judge’s training data are graded least reliably.

References

- Stella Biderman, USVSN Sai Prashanth, Lintang Sutawika, Hailey Schoelkopf, Quentin Anthony, Shivanshu Purohit, and Edward Raff. Emergent and predictable memorization in large language models. *arXiv preprint arXiv:2304.11158*, 2023a.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, 2023b.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few shot learners. In *Advances in Neural Information Processing Systems*, 2020. arXiv:2005.14165.
- Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramer, and Chiyuan Zhang. Quantifying memorization across neural language models. In *International Conference on Learning Representations*, 2023. arXiv:2202.07646.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Alvan R Feinstein and Domenic V Cicchetti. High agreement but low kappa: I. the problems of two paradoxes. *Journal of Clinical Epidemiology*, 43(6):543–549, 1990.
- Jiang Liu, Jialian Wu, Xiaodong Yu, Yusheng Su, Prakamya Mishra, Gowtham Ramesh, Sudhanshu Ranjan, Chaitanya Manem, Ximeng Sun, Ze Wang, Pratik Prabhanjan Brahma, Zicheng Liu, and Emad Barsoum. Instella: Fully open language models with stellar performance. *arXiv preprint arXiv:2511.10628*, 2025.
- Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. Detecting pretraining data from large language models. In *International Conference on Learning Representations*, 2024. arXiv:2310.16789.
- Shuo Yang, Wei-Lin Chiang, Lianmin Zheng, Joseph E Gonzalez, and Ion Stoica. Rethinking benchmark and contamination for language models with rephrased samples. *arXiv preprint arXiv:2311.04850*, 2023.
- Hugh Zhang, Jeff Da, Dean Lee, Vaughn Robinson, Catherine Wu, Will Song, Tiffany Zhao, Pranav Raja, Dylan Slack, Qin Lyu, Sean Hendryx, Russell Kaplan, Michele Lunati, and Summer Yue. A careful examination of large language model performance on grade school arithmetic. *arXiv preprint arXiv:2405.00332*, 2024.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P Xing, Hao Zhang, Joseph E Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023. arXiv:2306.05685.

A Related work in full

Zheng et al. [Zheng et al., 2023] introduce the LLM as judge paradigm at scale and document position bias, verbosity bias and self preference, reporting agreement with human preference above 80 percent for strong judges. Their audits characterize the judge and the prompt; membership of the graded item in the judge’s corpus is outside reach for the closed models they study, which is the gap the present design addresses for one open family.

On the detection side, containment based overlap rules originate with Brown et al. [Brown et al., 2020], and membership inference from model confidence is developed by Carlini et al. [Carlini et al., 2023] and Shi et al. [Shi et al., 2024]. Yang et al. [Yang et al., 2023] show that rephrasing defeats overlap based detection outright, which is why a study that needs a trustworthy treatment label cannot rely on an estimated one.

Two lines of work establish the value of released intermediate checkpoints. Biderman et al. [Biderman et al., 2023b] release the Pythia suite specifically to make training dynamics analyzable, and Biderman et al. [Biderman et al., 2023a] use such checkpoints to study when memorization emerges during training. The present work applies the same logic to an evaluator rather than to a generator.

B Containment method and calibration

Item text is normalized to lowercase alphanumeric tokens and decomposed into every window of 13 consecutive tokens. Containment against a corpus is the fraction of an item’s windows found in it, computed in one streaming pass with an inverted index from window to source item, so cost is linear in corpus size and independent of how many item sets are indexed together. Because the corpus is generated only from the GSM8K training split, every test item is a verified negative: the any match rule flags 3 of 1000, a rate of 0.3 percent, and the adopted threshold flags none.

Identifier namespacing is load bearing. GSM8K train and test share an index scheme, so loading both without namespacing collides items and produces containment values above unity, which is impossible for a fraction. Anyone reproducing this design will meet the same trap.

C Reproducibility controls

Several controls are load bearing rather than incidental. The transformers version is pinned exactly, since a version range places different checkpoints on different minor versions and that difference enters the estimate as though it were a model difference. Batch size is fixed across all generation, because greedy decoding in reduced precision is sensitive to padding and a batch size change alters tie breaking. Variants are seeded per pair of item and perturbation type, so variant text is stable regardless of call order and generations stay valid when the item set changes. Every generation batch is flushed to disk and resume is keyed on item identifier, so an interrupted run continues instead of restarting.

All generation records, containment values, judge verdicts and analysis code are released for audit at a public repository and dataset mirror, both anonymized for review. Random seeds are fixed across arm selection, variant construction and every bootstrap. An assistant helped implement generation, containment and analysis code under the direction of the authors, who verified every reported number against the underlying verdict files; the assistant did not design the conditions or choose which results to report.